Linear Algebra - 24DS

Consolidated Assignment No. 1

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There are two types of questions; one "Mechanical" which requires you to do computations and other "Conceptual", which requires you to respond question logically using theorem and logical implications. Attempt accordingly.

I RREF and Determinant

1. **(Mechanical)** Consider the matrix

$$A = \begin{bmatrix} 2 & 1 & 3 \\ 1 & 2 & 1 \\ 3 & 4 & 5 \end{bmatrix}.$$

   Find the determinant of A by:

   (a) Reducing A to an **upper triangular matrix**.

   (b) Reducing A to **RREF** (row-reduced echelon form).

   (c) Show that the determinant is the same in both methods.

2. **(Mechanical)** Find inverse of matrix A in Q 1.

3. **(Mechanical/Conceptual)** Show identities without evluation determinants; you may get help from row operations.

   (a)

$$\begin{vmatrix} a_1 & b_1 & a_1+b_1+c_1 \\ a_2 & b_2 & a_2+b_2+c_2 \\ a_3 & b_3 & a_3+b_3+c_3 \end{vmatrix} = \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}$$

   (b)

$$\begin{vmatrix} a_1+b_1t & a_2+b_2t & a_3+b_3t \\ a_1+b_1t^2 & a_2+b_2t^2 & a_3+b_3t^2 \\ c_1 & c_2 & c_3 \end{vmatrix} = (1-t^2)\begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix}$$

(c)

$$\begin{vmatrix} a_1+b_1 & a_1-b_1 & c_1 \\ a_2+b_2 & a_2-b_2 & c_2 \\ a_3+b_3 & a_3-b_3 & c_3 \end{vmatrix} = -2 \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}$$

(d)

$$\begin{vmatrix} a_1+ta_1 & c_1+rb_1+sa_1 \\ a_2+ta_2 & c_2+rb_2+sa_2 \\ a_3+ta_3 & c_3+rb_3+sa_3 \end{vmatrix} = \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}$$

II Solving linear sytem using row operations using RREF(necessary)

4. **(Mechanical)** A cybersecurity firm monitors network traffic through three routers. The total data throughput (in gigabits per second) through the three routers satisfies the system:

$$\begin{cases} 2x - y + 3z = 12 \\ x + 4y - 2z = 5 \\ 3x - 2y + z = 8 \end{cases}$$

where x, y, and z represent the throughput of Routers A, B, and C respectively. However, a distributed denial-of-service (DDoS) attack adds an additional constant traffic load to each equation, transforming it into:

$$\begin{cases} 2x - y + 3z = 12 + a \\ x + 4y - 2z = 5 + b \\ 3x - 2y + z = 8 + c \end{cases}$$

If the attack vector $(a, b, c) = (4, 1, 3)$, find the original traffic distribution (x, y, z) that satisfies both the original and attacked systems simultaneously. What does this tell you about the relationship between the two systems?

5. **(Mechanical)** In cryptography for a linear feedback shift register (LFSR) analysis, the system is:

$$\begin{cases} x_1 + x_3 + x_4 = 1 \\ x_1 + x_2 + x_4 = 0 \\ x_2 + x_3 + x_4 = 1 \\ x_1 + x_2 + x_3 = 0 \end{cases}$$

(modulo 2 arithmetic). Write the augmented matrix over $\mathbb{F}_2$ and reduce to RREF to find the initial state vector.

6. **(Mechanical)** In image processing, a color image is represented as a combination of red, green, and blue intensity values. A filtering algorithm applies three different filters

to an image, producing the following relationships between the original pixel intensities (r, g, b) and the filtered outputs:

$$\begin{cases} 2r + 3g - b = 120 \\ r - 2g + 4b = 85 \\ 3r + g + 2b = 140 \end{cases}$$

After processing, you discover that the blue channel sensor was miscalibrated, adding an extra 10 units to every equation involving b. Find the true original intensities if the miscalibration is corrected.

7. **(Mechanical)** In cryptography, a 4-bit message (x_1, x_2, x_3, x_4) is encrypted using a linear transformation that yields zero output for certain messages. These "null messages" satisfy:

$$\begin{cases} x_1 + 2x_2 - x_3 + x_4 = 0 \\ 2x_1 - x_2 + 3x_3 - x_4 = 0 \\ x_1 + x_2 - 2x_3 + 2x_4 = 0 \\ 3x_1 - x_2 + x_3 - x_4 = 0 \end{cases}$$

Find all possible null messages. If the encryption scheme is insecure when the dimension of the solution space exceeds 1, is this scheme secure?

8. **(Mechanical)** In computer graphics, a 3D object's vertices undergo transformation (x, y, z, w) in homogeneous coordinates. Fixed points of a certain projection satisfy:

$$\begin{cases} x - y + z - w = 0 \\ 2x + y - z + w = 0 \\ x + 3y + z - 2w = 0 \\ -x + 2y + 3z + w = 0 \end{cases}$$

Find all fixed points. Describe the subspace they form in $\mathbb{R}^4$.

9. **(Mechanical)** Solve the following homogeneous linear system by reducing it to **row echelon form**:

$$\begin{aligned} w + x + 2y + 3z &= 0 \\ x + y + 2z &= 0 \\ 2w + 2x + 6y + 8z &= 0 \\ 2x + 2y + 4z &= 0 \end{aligned}$$

Does the system have **infinitely many solutions**? If yes:

(a) Write two different solutions.

(b) Show that the sum of these two solutions is again a solution of the system.

Also determine the **rank of the coefficient matrix**.

10. **(Conceptual)** Write the coefficient matrix of the system into a (possibly different) **row echelon form** using a sequence of row operations **different from those used in Question 9**.

III Linear Combinations

1. In computer vision, three feature vectors extracted from an image are:

$$\mathbf{f}_1 = (2, 1, 3), \quad \mathbf{f}_2 = (-1, 2, 1), \quad \mathbf{f}_3 = (4, -1, 2)$$

   A new image produces feature vector $\mathbf{g} = (7, 3, 8)$. Can $\mathbf{g}$ be expressed as a linear combination of $\mathbf{f}_1, \mathbf{f}_2, \mathbf{f}_3$? If so, find the coefficients. What does this indicate about the features' ability to represent new images?

2. Three color palettes in a graphic design tool are represented as RGB vectors:

$$\mathbf{p}_1 = (255, 0, 0), \quad \mathbf{p}_2 = (0, 255, 0), \quad \mathbf{p}_3 = (0, 0, 255)$$

   A designer wants to create a new color $\mathbf{c} = (128, 128, 255)$ by mixing these palettes. Find the mixing ratios. Is this combination unique?

3. In audio processing, three sound waves are represented by amplitude vectors:

$$\mathbf{w}_1 = (1, 2, -1), \quad \mathbf{w}_2 = (2, -1, 3), \quad \mathbf{w}_3 = (-1, 3, 2)$$

   A composite wave $\mathbf{s} = (4, 5, 5)$ is detected. Decompose $\mathbf{s}$ into a linear combination of $\mathbf{w}_1, \mathbf{w}_2, \mathbf{w}_3$. Interpret the coefficients as volumes of each source wave.

4. Three basis vectors in a 3D rendering engine are:

$$\mathbf{b}_1 = (1, 1, 0), \quad \mathbf{b}_2 = (1, 0, 1), \quad \mathbf{b}_3 = (0, 1, 1)$$

   A vertex is located at $\mathbf{v} = (3, 2, 2)$ in standard coordinates. Express $\mathbf{v}$ in terms of the basis $\{\mathbf{b}_1, \mathbf{b}_2, \mathbf{b}_3\}$. If the basis is orthogonalized, would the coordinates change?

IV Spaning/Generating vectors

5. A robot arm moves in 3D space with three joints. The reachable positions are all linear combinations of:

$$\mathbf{a}_1 = (2, 1, 0), \quad \mathbf{a}_2 = (1, 2, 1), \quad \mathbf{a}_3 = (3, 3, 1)$$

   Does the set $\{\mathbf{a}_1, \mathbf{a}_2, \mathbf{a}_3\}$ span $\mathbb{R}^3$? If not, describe the subspace of reachable positions. Can the robot reach position $(1, 1, 2)$?

6. In data science, three feature vectors from a dataset are:

$$\mathbf{d}_1 = (1, 2, 3), \quad \mathbf{d}_2 = (2, 3, 4), \quad \mathbf{d}_3 = (3, 4, 5)$$

   Do these vectors span $\mathbb{R}^3$? If not, what is the dimension of their span? A new data point $(4, 5, 6)$ is collected. Is it in the span? What does this imply about adding redundant features?

7. Three force vectors act on a physics simulation:

$$\mathbf{f}_1 = (1, 0, 1), \quad \mathbf{f}_2 = (0, 1, 1), \quad \mathbf{f}_3 = (1, 1, 0)$$

Do these forces span all possible 3D force directions? Find the set of all force directions that cannot be achieved by combining these three forces.

8. A neural network's hidden layer produces outputs that are linear combinations of:

$$\mathbf{h}_1 = (1, -1, 2), \quad \mathbf{h}_2 = (2, 1, -1), \quad \mathbf{h}_3 = (3, 0, 1)$$

Determine whether $\{\mathbf{h}_1, \mathbf{h}_2, \mathbf{h}_3\}$ spans $\mathbb{R}^3$. If not, find a basis for their span. A new sample produces $(4, -1, 4)$. Can the network represent this output?

V Linear Independence of Vectors

9. In cryptography, three encryption keys are represented as vectors:

$$\mathbf{k}_1 = (1, 2, 3), \quad \mathbf{k}_2 = (2, 3, 1), \quad \mathbf{k}_3 = (3, 1, 2)$$

Are these keys linearly independent? If they are dependent, find a nontrivial combination that yields zero. How does this affect the security of the encryption scheme?

10. Three network traffic patterns are recorded as vectors of packet counts through three routers:

$$\mathbf{t}_1 = (150, 200, 100), \quad \mathbf{t}_2 = (300, 400, 200), \quad \mathbf{t}_3 = (75, 100, 50)$$

Determine whether these traffic patterns are linearly independent. If they are dependent, explain which pattern can be expressed as a combination of the others and what this means for network monitoring.

11. In machine learning, three feature vectors from a dataset are:

$$\mathbf{x}_1 = (1, 0, 1), \quad \mathbf{x}_2 = (0, 1, 1), \quad \mathbf{x}_3 = (1, 1, 0)$$

Check for linear independence. If they are independent, can they form a basis for $\mathbb{R}^3$? If not, find the dimension of their span and a basis for it.

12. Three color vectors in an RGB color space are

$$\mathbf{c}_1 = (255, 128, 0), \quad \mathbf{c}_2 = (128, 255, 0), \quad \mathbf{c}_3 = (0, 128, 255)$$

Test these vectors for linear independence. If they are dependent, find the relationship. Would removing one vector reduce the color gamut? Explain.

VI Bases and change of bases

1. **(Mechanical)** Show that the set $\left\{ \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 \\ 1 \end{pmatrix} \right\}$ is a basis for $\mathbb{R}^2$. Express $\begin{pmatrix} 3 \\ 2 \end{pmatrix}$ in this basis.

2. **(Mechanical)** In $\mathbb{R}^2$, let $\mathcal{B} = \{(1, 2), (2, 1)\}$ and $\mathcal{C} = \{(1, 0), (0, 1)\}$. Find the change-of-basis matrix from $\mathcal{B}$ to $\mathcal{C}$ and from $\mathcal{C}$ to $\mathcal{B}$.

3. **(Conceptual)** If $\mathcal{B} = \{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$ is a basis for $\mathbb{R}^3$, is $\{\mathbf{v}_1 + \mathbf{v}_2, \mathbf{v}_2 + \mathbf{v}_3, \mathbf{v}_3 + \mathbf{v}_1\}$ necessarily a basis? Prove or give a counterexample.

4. **(Mechanical)** Let $\mathcal{B} = \left\{ \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} \right\}$. Find the change-of-basis matrix $P_{\mathcal{B} \leftarrow \mathbf{B}}$ where $\mathbf{B}$ is the standard basis. Then find coordinates of $\begin{pmatrix} 2 \\ 3 \\ 4 \end{pmatrix}$ in basis $\mathcal{B}$.

5. **(Conceptual)** Explain why any set of 4 vectors in $\mathbb{R}^3$ cannot be linearly independent. Use the definition of basis.

6. **(Conceptual)** Suppose P is the change-of-basis matrix from $\mathcal{B}$ to $\mathcal{C}$. What is the relationship between P and the change-of-basis matrix from $\mathcal{C}$ to $\mathcal{B}$? Prove it.

7. **(Mechanical)** Determine if $\left\{ \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 2 \end{pmatrix}, \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} \right\}$ is a basis for $\mathbb{R}^3$. If so, find the coordinates of $\begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$ in this basis.

8. **(Conceptual)** Can a basis for $\mathbb{R}^4$ contain the zero vector? Why or why not?

VII Subspaces, thier basis and dimension (18 questions)

1. **($\mathbb{R}^3$ - Line through origin)** Determine whether

$$W_1 = \left\{ \begin{bmatrix} x \\ y \\ z \end{bmatrix} \in \mathbb{R}^3 \;\middle|\; x = 2t,\ y = -t,\ z = 3t,\ t \in \mathbb{R} \right\}$$

is a subspace of $\mathbb{R}^3$. If yes, find its dimension and geometric description.

2. **($\mathbb{R}^3$ - Plane through origin)** Check if

$$W_2 = \left\{ \begin{bmatrix} x \\ y \\ z \end{bmatrix} \in \mathbb{R}^3 \;\middle|\; 2x - y + 3z = 0 \right\}$$

is a subspace of $\mathbb{R}^3$. Find a basis and state the dimension.

3. **($\mathbb{R}^3$ - Not containing zero)** Is

$$W_3 = \left\{ \begin{bmatrix} x \\ y \\ z \end{bmatrix} \in \mathbb{R}^3 \;\middle|\; x + y + z = 1 \right\}$$

a subspace of $\mathbb{R}^3$? Justify your answer.

4. **($\mathbb{R}^4$ - Hyperplane)** Show that

$$W_4 = \left\{ \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} \in \mathbb{R}^4 \;\middle|\; x_1 + x_2 - x_3 + 2x_4 = 0 \right\}$$

is a subspace of $\mathbb{R}^4$. Find its dimension and a basis.

5. **($\mathbb{R}^4$ - Span of vectors)** Let

$$W_5 = \operatorname{span} \left\{ \begin{bmatrix} 1 \\ 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} \right\}$$

Is W_5 a subspace of $\mathbb{R}^4$? Justify and find its dimension.

6. **(Conceptual)** Explain why every subspace of $\mathbb{R}^n$ must contain the zero vector. Give an example of a subset of $\mathbb{R}^3$ that contains the zero vector but is NOT a subspace.

1. **(Basis from span in $\mathbb{R}^3$)** Find a basis for

$$W = \operatorname{span}\left\{ \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}, \begin{bmatrix} 2 \\ 4 \\ 6 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} \right\}$$

2. **(Basis for a plane in $\mathbb{R}^3$)** Find a basis for the subspace

$$W = \left\{ \begin{bmatrix} x \\ y \\ z \end{bmatrix} \in \mathbb{R}^3 \;\middle|\; x - y + 2z = 0 \right\}$$

3. **(Basis for a line in $\mathbb{R}^4$)** Find a basis for

$$W = \left\{ \begin{bmatrix} 2t \\ -t \\ 3t \\ t \end{bmatrix} \;\middle|\; t \in \mathbb{R} \right\} \subseteq \mathbb{R}^4$$

4. **(Basis from column space)** Find a basis for the subspace spanned by the columns of

$$A = \begin{bmatrix} 1 & 2 & 3 & 4 \\ 2 & 4 & 6 & 8 \\ 1 & 0 & 1 & 2 \end{bmatrix}$$

5. **(Extending to a basis)** Extend the linearly independent set

$$\left\{ \begin{bmatrix} 1 \\ 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ 1 \\ 0 \end{bmatrix} \right\}$$

to a basis of $\mathbb{R}^4$.

6. **(Conceptual)** Can a set of 4 vectors in $\mathbb{R}^3$ form a basis for a subspace? If yes, what is the maximum possible dimension? Explain with an example.

1. **(Dimension of a plane in $\mathbb{R}^3$)** Find the dimension of

$$W = \left\{ \begin{bmatrix} x \\ y \\ z \end{bmatrix} \in \mathbb{R}^3 \,\middle|\, 3x - y + 2z = 0 \right\}$$

   Also, describe the geometric shape.

2. **(Dimension of span in $\mathbb{R}^4$)** Find the dimension of

$$W = \operatorname{span}\left\{ \begin{bmatrix} 1 \\ 2 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 2 \\ 4 \\ 0 \\ 2 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ 1 \\ 1 \end{bmatrix} \right\}$$

3. **(Dimension of solution space in $\mathbb{R}^4$)** Find the dimension of the solution space of

$$x_1 + x_2 - x_3 = 0, \quad 2x_1 - x_2 + x_4 = 0$$

   (in $\mathbb{R}^4$).

4. **(Dimension of intersection)** Find the dimension of $W_1 \cap W_2$ where

$$W_1 = \operatorname{span}\{(1,0,0),(0,1,0)\}, \quad W_2 = \operatorname{span}\{(1,1,0),(0,0,1)\}$$

   in $\mathbb{R}^3$.

5. **(Dimension comparison)** Let W_1 be a plane through origin in $\mathbb{R}^3$ and W_2 be a line through origin in $\mathbb{R}^3$ not contained in W_1. What is $\dim(W_1 \cap W_2)$? Justify.

6. **(Conceptual)** Is it possible for a subspace of $\mathbb{R}^4$ to have dimension 5? Why or why not? Explain the relationship between dimension and the ambient space.